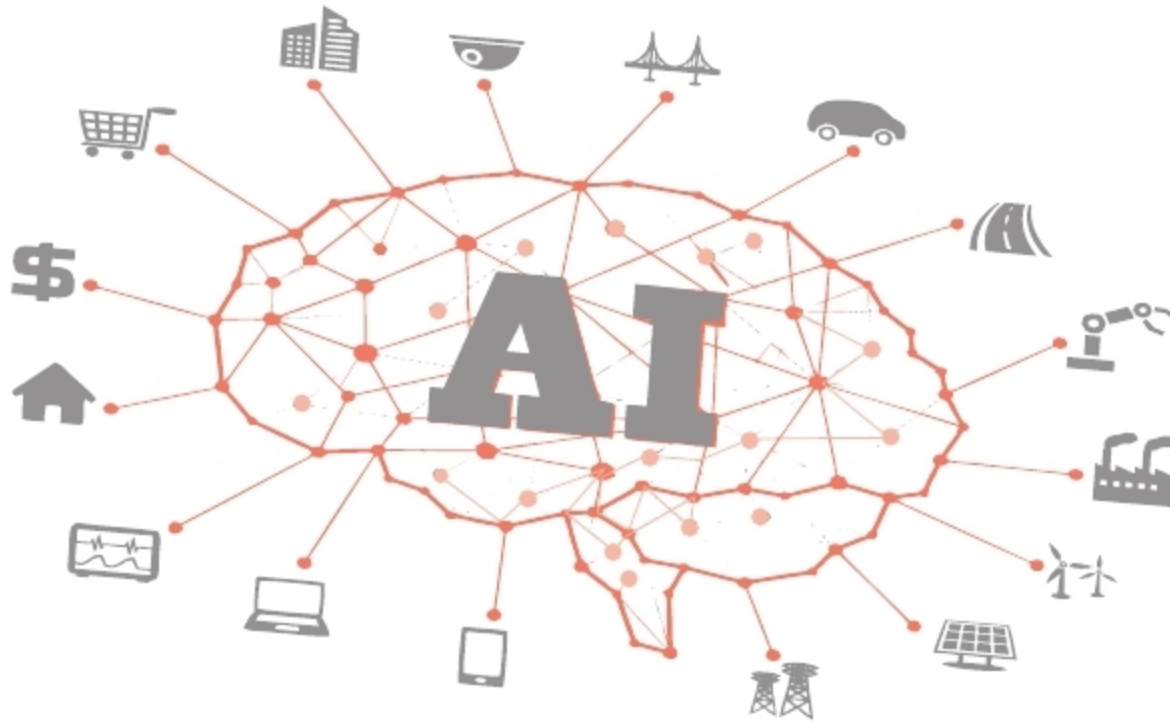


AI and the IoT: Leveraging Digital Disruption

This article discusses the combined applications of artificial intelligence (AI) and the Internet of Things (IoT) technologies, which are set to transform modern industry and lead to a bright and promising future.



Imagine a piece of wood doing amazing things using artificial intelligence (AI) and the Internet of Things (IoT)! This is exactly what mui smart wooden plank (Figure 1) does. It is an interactive wood panel for smart home applications, created by a Japanese company and showcased recently at Consumer Electronics Show (CES) 2019, Las Vegas.

mui is an Internet-connected plank of wood with built-in microphone, touch-sensitive interface and Google Assistant. It can display information including weather, and also control Internet-connected devices and smart assistants. Its design interface includes elements of AI and the IoT. From metal (conductor) to semiconductor, and now to wood! Who knows, in the future, with the power of technologies such as AI and the IoT, your chairs and tables could become smart enough to start tracking your daily routines and activities.

Currently, AI and the IoT are the buzzwords in the industry. These technologies fascinate many researchers and industries across the globe. A combination of these two technologies is revolutionary, and is set to transform the landscape of modern industry. The growth of both technologies will impact the industry, including its structures, transactions and practices.

These technologies will impact future businesses and help reduce global poverty, solve water shortage problems and improve agriculture productions. These will greatly benefit emerging countries like India and Africa. A smart connectivity ecosystem coupled with AI will be a disruptive force that will change the world in a big way. Almost everything will be smarter, including self-driving cars, smart homes and smart cities.

This article is not about the merits and demerits of AI and the IoT; rather, it is about the combined applications of these technologies, which are set to

have a bright and promising future in transforming the modern industry. AI and the IoT work hand-in-hand in areas such as Big Data, edge computing, smart homes, automotive, healthcare, agriculture, manufacturing, wearables and military, among others.

Convergence of AI and the IoT

Before delving into AI's and the IoT's specific applications, let us first see the main differences between them and then their point of intersection.

Broadly speaking, AI and the IoT are two different types of technologies. AI emphasises on making intelligent machines work and react like human beings, including speech recognition, planning, problem solving and learning. It involves programming intelligence or making computers behave like humans. AI-powered devices try to imitate the natural intelligence of humans, mimicking the cognitive functions that humans use to perform tasks.

On the other hand, the IoT is the interworking of physical devices like sensors and actuators, which are able to communicate among themselves or with the external environment including cars, home appliances and human beings. The IoT involves extending Internet connectivity beyond standard devices, such as computers and smartphones, to any range of non-Internet-enabled physical devices and everyday objects. As per Wikipedia, the number of IoT devices increased 31 per cent year-over-year to 8.4 billion in 2017, and it is estimated that there will be 30 billion such devices by 2020.

The IoT can be found in wearable devices, smart homes, various modes of transportation, agriculture and